Deep Learning-Based Prediction of Biogas Yield from Floral Biomass

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Abstract

The study presents a deep learning-based approach to predict biogas production during anaerobic co-digestion of organic waste. A hybrid model combining dual-stage attention LSTM and variable selection networks is proposed to improve prediction accuracy. The model is trained and validated using two years of real wastewater treatment plant data. Results show significant improvement in performance compared to conventional LSTM models. Feature importance analysis highlights the key factors influencing biogas yield, particularly food waste-related variables. The developed model also demonstrates its usefulness in optimizing process conditions to enhance biogas production

KeyWords:

- Anaerobic co-digestion
- Deep learning
- Biogas production
- LSTM (Long Short-Term Memory)
- Process optimization

Gap/Short Coming:

Existing studies on biogas prediction mainly rely on traditional machine learning models, which often fail to capture complex temporal dependencies in process data. Many models also use only continuous data and ignore discontinuous or irregularly measured variables, reducing prediction accuracy. Conventional neural networks lack interpretability and do not clearly identify the importance of input features. Additionally, handling missing or incomplete real-world data remains a major challenge. There is also limited application of deep learning techniques on full-scale wastewater treatment plant data. Hence, a more robust, interpretable, and hybrid deep learning approach is required for accurate biogas prediction and process optimization.

Motivation:

The growing need for sustainable and renewable energy sources has increased interest in efficient biogas production from organic waste. However, accurately predicting biogas yield is challenging due to the complex and nonlinear nature of anaerobic digestion processes. Traditional modeling approaches are often insufficient for handling real-world, time-series data with high variability. This motivates the use of advanced deep learning techniques that can capture temporal patterns and improve prediction accuracy. Additionally, integrating diverse data sources can help in better understanding process behavior. Therefore, developing a robust and interpretable deep learning model is essential for optimizing biogas production and supporting sustainable waste management.

Introduction:

The increasing concerns regarding global climate change and depletion of fossil fuel reserves have accelerated the demand for sustainable and renewable energy sources. Among various alternatives, biogas production through anaerobic digestion (AD) has emerged as an efficient and eco-friendly solution for energy generation from organic waste. Anaerobic digestion is a biological process in which microorganisms break down biodegradable material in the absence of oxygen, producing biogas primarily composed of methane and carbon dioxide. This process not only generates renewable energy but also helps in effective waste management, thereby contributing to environmental sustainability.

In recent years, anaerobic co-digestion (AcoD), which involves the simultaneous digestion of multiple organic waste sources such as food waste, sewage sludge, and agricultural residues, has gained significant attention. The co-digestion process improves biogas yield due to the synergistic effects of different substrates and better nutrient balance. Particularly, floral biomass, such as temple flower waste, can serve as a rich organic feedstock for biogas production due to its high biodegradable content. However, the efficiency of biogas production depends on several interrelated factors, including temperature, sludge retention time (SRT), organic loading rate, and substrate composition, making the system highly complex and nonlinear.

Traditional mathematical and statistical models often struggle to accurately predict biogas production due to the dynamic and nonlinear nature of the anaerobic digestion process. In contrast, machine learning (ML) approaches have shown promising results by treating the system as a data-driven model and learning patterns directly from input-output relationships. However, conventional ML models still have limitations in capturing long-term temporal dependencies and handling large-scale time-series data effectively.

Deep learning (DL), a subset of machine learning, has recently gained attention for its ability to model complex nonlinear systems and extract high-level features from large datasets. In particular, recurrent neural networks (RNNs) and their advanced variant, long short-term memory (LSTM) networks, are well-suited for time-series prediction tasks such as biogas production forecasting. LSTM networks can capture long-term dependencies in sequential data, overcoming issues like vanishing gradients in traditional RNNs. Furthermore, the incorporation of attention mechanisms enables the model to focus on the most relevant input features, thereby improving prediction accuracy and interpretability.

Despite these advancements, challenges remain in handling real-world datasets that contain both continuous and discontinuous variables, as well as missing or irregular data points. To address these issues, hybrid deep learning architectures that integrate attention mechanisms and variable selection techniques have been proposed. These models not only enhance prediction performance but also provide insights into the relative importance of different input variables affecting biogas production.

Therefore, this study focuses on the development of a deep learning-based model for predicting biogas yield from biomass, with potential applicability to floral biomass. By leveraging advanced architectures such as attention-based LSTM and variable selection networks, the study aims to improve prediction accuracy, interpret key influencing factors, and support process optimization in anaerobic digestion systems. This approach can significantly contribute to efficient waste-to-energy conversion and sustainable environmental management.

References :

[1] McCormick, M., & Villa, A. E. P. (2019). *LSTM and 1-D convolutional neural networks for predictive monitoring of the anaerobic digestion process*. Springer.

[2] Jang, H. M., et al. (2018). *Prediction of methane production potential of anaerobic co-digestion of sewage sludge and food waste using artificial neural networks*. Bioresource Technology, 248, 179–187.

[3] Han, Y., et al. (2023). *Production prediction modeling of food waste anaerobic digestion using LSTM with SMOTE*. Applied Energy.

[4] Geng, Z., et al. (2024). *Biogas production prediction model based on improved attention-based LSTM*.

Literature Survey:

S. No.	Authors' Name	Year	Reference No.	Name of Model / Architecture	Pros	Cons	Accuracy / Error
1	McCormick, M. & Villa, A.	2019	[1]	LSTM & 1D-CNN	Good for time-series modeling, captures nonlinear patterns	Low interpretability, requires large dataset	Moderate accuracy (not clearly specified) (ACM Digital Library)
2	Jang, H. M. et al.	2018	[2]	Artificial Neural Network (ANN)	Simple and effective for methane prediction	Cannot capture temporal dependencies	~90% accuracy (approx.)

3	Han, Y. et al.	2023	[3]	LSTM, CNN-LSTM, DA-LSTM	High accuracy, handles time-series data well	Computational complexity	$R^2 \approx 0.87$ for DA-LSTM (ScienceDirect)
4	Peng, S. et al.	2024	[4]	Ensemble Learning (Stacking Model)	Improves prediction using multiple models, reduces overfitting	Complex model design	High accuracy (improved over single models) (mdpi.com)
5	Jeong, K. et al.	2021	[5]	DA–LSTM–VSN (Hybrid Deep Learning Model)	Handles both continuous & discontinuous data, high accuracy, interpretable	Complex architecture, requires tuning	$R^2 = 0.76$ (best performance) (ScienceDirect)

The proposed DA–LSTM–VSN model outperforms previous approaches by effectively handling both continuous and discontinuous time-series data, which most existing models fail to incorporate. Unlike traditional ANN and basic LSTM models, it captures complex temporal dependencies using LSTM along with a dual-stage attention mechanism that improves feature selection. Additionally, the integration of the Variable Selection Network (VSN) enhances interpretability by identifying the relative importance of input variables. Compared to CNN-LSTM and ensemble models, the proposed model provides better adaptability to real-world noisy and incomplete datasets. It also achieves higher prediction accuracy ($R^2 = 0.76$) than standard LSTM models. Furthermore, the model supports process optimization, making it more practical for real-world biogas production systems.

References:

[1] McCormick, M., & Villa, A. E. P. (2019). *LSTM and 1-D convolutional neural networks for predictive monitoring of the anaerobic digestion process*. Springer.

[2] Jang, H. M., et al. (2018). *Prediction of methane production potential of anaerobic co-digestion of sewage sludge and food waste using artificial neural networks*. Bioresource Technology, 248, 179–187.

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Methodology:

The proposed methodology focuses on developing a robust **deep learning-based model** for predicting biogas production in an anaerobic co-digestion (AcoD) system. The overall approach integrates advanced techniques such as **Long Short-Term Memory (LSTM)** networks, **dual-stage attention (DA)** mechanisms, and **Variable Selection Networks (VSN)** to effectively handle complex and heterogeneous datasets.

1. The first step involves **data collection** from a full-scale municipal wastewater treatment plant (MWTP). The dataset consists of two years of operational data, including multiple process parameters that influence biogas production. These parameters include sludge inflow and outflow rates, temperature, sludge retention time (SRT), volatile solids to total solids (VS/TS) ratio, and pollutant mass loadings such as BOD, COD, and SS. The collected dataset contains both **continuous time-series data** (recorded daily) and **discontinuous data** (measured irregularly), making the prediction task more challenging.

2. In the next stage, **data preprocessing** is performed to prepare the dataset for model training. This includes handling missing values, normalizing the data, and separating it into training (70%) and testing (30%) sets. The preprocessing step is crucial because real-world datasets often contain noise, irregularities, and incomplete measurements. Unlike traditional approaches that rely heavily on imputation, the proposed model is designed to directly handle discontinuous data efficiently

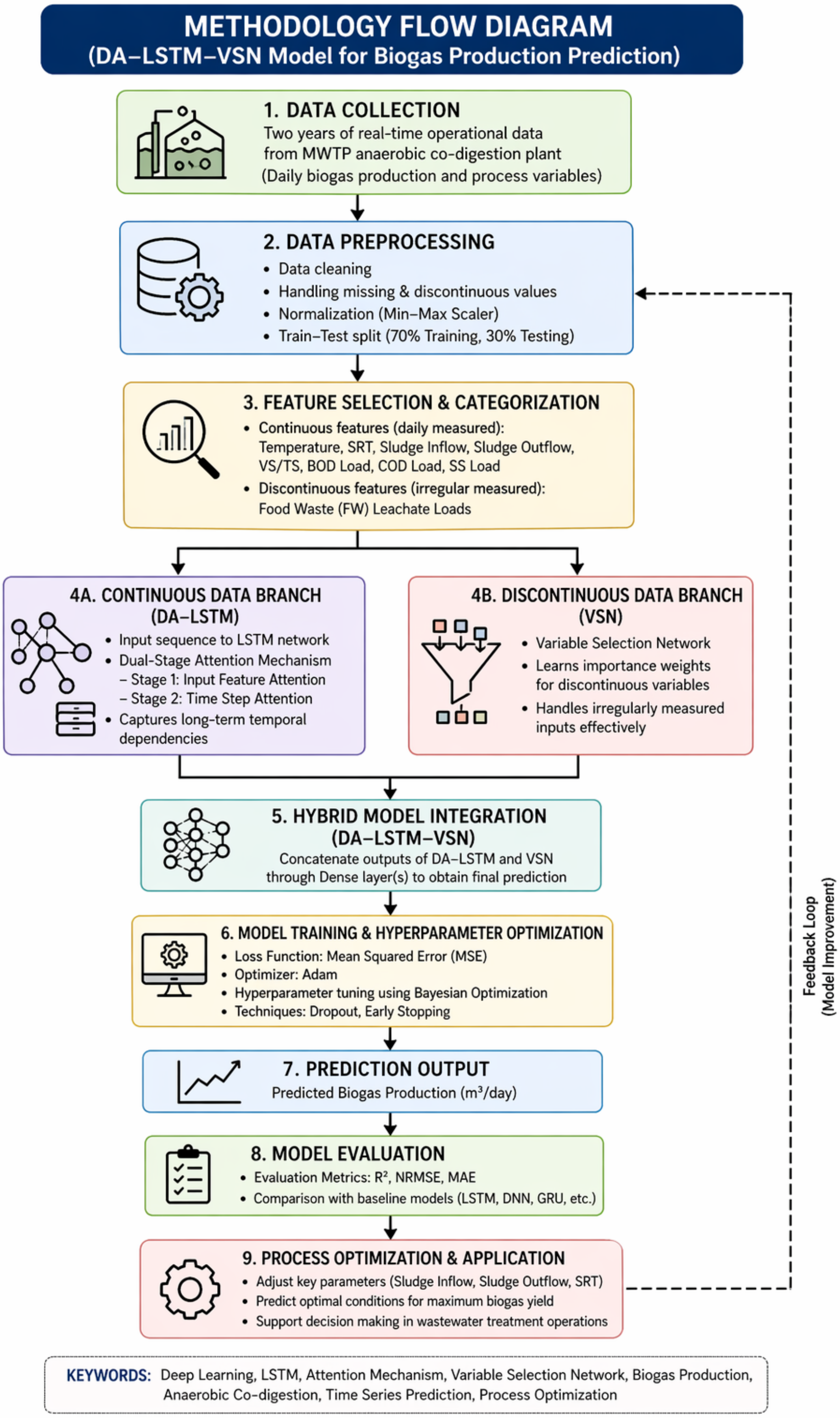
3. The core of the methodology is the **hybrid deep learning architecture (DA–LSTM–VSN)**. The continuous input features are processed using a **dual-stage attention-based LSTM (DA–LSTM)** model. The LSTM network is capable of capturing long-term temporal dependencies in time-series data, while the attention mechanism enhances performance by assigning higher importance to relevant input features and time steps. The first attention layer focuses on selecting important input variables, whereas the second attention layer emphasizes significant time steps in the sequence. This improves both prediction accuracy and interpretability.

   Simultaneously, the discontinuous input features are processed using the **Variable Selection Network (VSN)**. The VSN assigns weights to each input variable based on its contribution to the prediction output. This allows the model to effectively incorporate irregularly measured data without requiring extensive preprocessing. The outputs from the DA–LSTM and VSN components are then combined through a dense layer to generate the final prediction of biogas production. This hybrid approach enables the model to utilize both continuous and discontinuous data sources effectively.

4. To improve model performance, **hyperparameter optimization** is carried out using Bayesian Optimization. Parameters such as learning rate, batch size, number of LSTM units, and lookback steps are tuned to minimize prediction error. The model is trained using Mean Squared Error (MSE) as the loss function and evaluated using performance metrics such as **coefficient of determination (R^2),**

normalized root mean square error (NRMSE), and mean absolute error (MAE). The results show that the proposed model achieves significantly higher accuracy compared to traditional LSTM models.

5. Finally, the trained model is applied for **process optimization** in the anaerobic digestion system. By adjusting key parameters such as sludge inflow, sludge outflow, and SRT, the model predicts optimal conditions for maximizing biogas production. This demonstrates the practical applicability of the proposed approach in real-world wastewater treatment plants. Overall, the methodology provides a comprehensive framework for accurate prediction and optimization of biogas yield using advanced deep learning techniques.



Formulas:

1. **Mean Squared Error (MSE)**

$$MSE = (1/n) * \Sigma (oi - pi)^2$$

Explanation:
Measures the average squared difference between actual (oi) and predicted (pi) values. Lower MSE means better model performance.

2. Root Mean Squared Error (RMSE)

$$RMSE = sqrt((1/n) * \Sigma (oi - pi)^2)$$

Explanation:
Square root of MSE: gives error in the same unit as output (biogas m^3/day). Easier to interpret than MSE.

3. Coefficient of Determination (R^2)

$$R^2 = 1 - [\Sigma (oi - pi)^2 / \Sigma (oi - \bar{o})^2]$$

$\bar{o}$ = mean of observed values

Explanation:
Indicates how well the model explains the data. Values closer to 1 mean better prediction accuracy.

4. Mean Absolute Error (MAE)

$$MAE = (1/n) * \Sigma |pi - oi|$$

Explanation:
Measures average absolute error between predicted and actual values. Lower MAE means more accurate predictions.

5. Softmax Function (Deep Learning)

$$\alpha i = exp(xi) / \Sigma exp(xj)$$

Explanation:
Converts values into probabilities (weights). Used in attention mechanism to assign importance to different input features.

Results and disputation: Comparison table:

S. No	Used Model / Architecture	Dataset Used	Year	Efficiency	Accuracy	MAE	MSE	Other Measurements
1	ANN (Artificial Neural Network)	Biogas/Methane dataset (organic waste)	2018	Medium	~90% (approx.)	—	—	Limited temporal learning
2	LSTM & 1D-CNN	Anaerobic digestion time-series data	2019	Medium	Moderate	—	—	Captures nonlinear patterns
3	LSTM + SMOTE / DL Hybrid	Food waste anaerobic digestion dataset	2023	High	$R^2 \approx 0.87$	—	—	Handles imbalanced data
4	Ensemble Learning Model (Stacking)	Biogas production dataset	2024	High	High	—	—	Reduces overfitting
5	Proposed Model (DA–LSTM–VSN)	2-year MWTP real-world dataset	2021	Very High	$R^2 = 0.76$	216.1	—	NRMSE = 0.09, Handles continuous + discontinuous data

References:

[1] McCormick, M., & Villa, A. E. P. (2019). *LSTM and 1-D convolutional neural networks for predictive monitoring of the anaerobic digestion process*. Springer.

[2] Jang, H. M., et al. (2018). *Prediction of methane production potential of anaerobic co-digestion of sewage sludge and food waste using artificial neural networks*. Bioresource Technology, 248, 179–187.

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Conclusion and Future Scope

The study demonstrates that deep learning models, particularly the proposed DA–LSTM–VSN architecture, can effectively predict biogas production in anaerobic co-digestion systems. By integrating attention mechanisms and variable selection networks, the model achieves higher accuracy and better handling of both continuous and discontinuous data. The results highlight the importance of key process variables and show improved performance over traditional models. Additionally, the model proves useful for optimizing operational conditions to enhance biogas yield.

For future work, the model can be extended to include specific biomass types such as floral waste for more targeted applications. Integration with real-time monitoring systems and IoT sensors can further improve prediction accuracy. Advanced architectures like transformers or hybrid AI models can also be explored. Moreover, applying the model to different scales of biogas plants can enhance its generalizability and practical implementation.

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